

Measurement-Based Quantum Kernel Method for Data Classification on a Reconfigurable Silicon Photonic Device

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Abstract We demonstrate a measurement-based quantum kernel estimation protocol on a reprogrammable silicon photonic chip. Using path-encoded four-qubit graph states we achieve 90 % classification accuracy. This provides a scalable, noise-resilient framework for quantum machine learning ©2026 The Author(s)

Introduction

Data classification is a cornerstone of modern machine learning, particularly for analyzing datasets with classes that are not linearly separable. For such tasks, support vector Machines (SVMs) are employed to map data into a higher-dimensional feature space using kernel functions. This generally allows us to define a hyperplane to classify data [1].

In recent years, quantum kernel estimation (QKE) has been explored as a promising candidate for achieving quantum advantage in specific learning tasks. Recent studies have applied this framework to quantum phase recognition [2], pattern matching [3], and to data with group structures [4]. Following this progress, most experimental demonstrations have relied on circuit-based quantum computing (CBQC) using superconducting [2, 4, 5], or trapped-ion [6] platforms. Photonic hardware has been explored for CBQC-QKE using continuous-variable states [7], unitary evolution of two boson Fock states [8], for sampling quantum similarity [9], and for performing swap tests [10]. They prove the possibility of implementing quantum algorithms on photonic platforms. However, these systems implement sequential quantum gates and suffer from limitations due to short decoherence times when scaling to larger qubit counts while maintaining high gate fidelities.

In this work, we overcome the barrier posed by decoherence time in CBQC by generating path-encoded graph states as a computational resource for measurement-based kernel mapping. To realize the mapping of input data to a quantum system, we propose a measurement-based approach to reconstructing the kernel matrix using multi-qubit graph states on a reprogrammable silicon photonic platform. This method has the potential to be both scalable with respect to the number of generated qubits and robust with respect to fidelity in reconstructing the kernel matrix. This hybrid approach uses quantum hardware for high-dimensional fea-

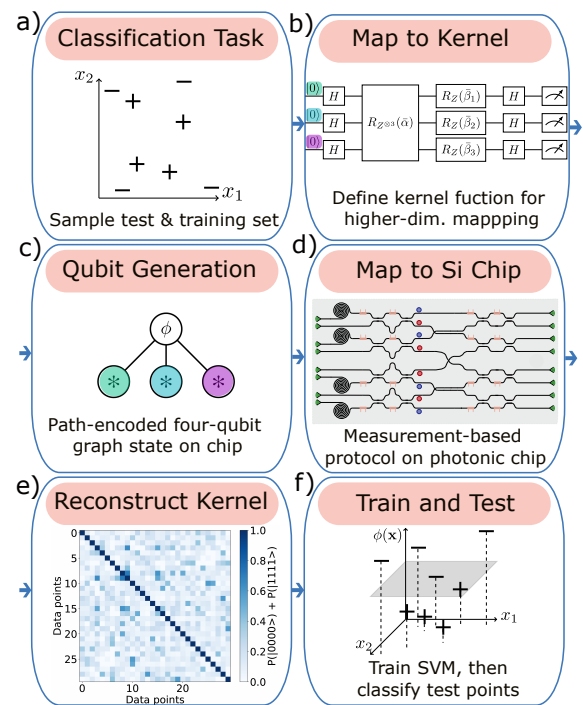


Fig. 1: Schematic workflow for the presented measurement-based quantum kernel classification protocol. a) Classification Task, b) Feature Mapping, c) Quantum Resource, d) Photonic Implementation, e) Matrix Reconstruction, f) Training and Testing

ture mapping while relying on classical optimization for final classification.

Quantum Kernel Protocol

The presented quantum kernel protocol builds on encoding data $\vec{x}$ in a quantum state $|\Phi(\vec{x})\rangle$, so that the corresponding Hilbert space $\mathcal{H}_\Phi$ can be accessed to increase the dimensionality of $\vec{x}$. Classical SVMs estimate pairwise distances between data points to compute similarity using the dot product. Translating this approach to our fidelity-based kernel matrix, the inner product $|\langle\Phi(\vec{x}_j)|\Phi(\vec{x}_i)\rangle|^2$ of two quantum states $|\Phi(\vec{x}_i)\rangle$ and $|\Phi(\vec{x}_j)\rangle$ is exploited [11].

The protocol for the measurement-based quantum kernel method presented in this paper is illustrated

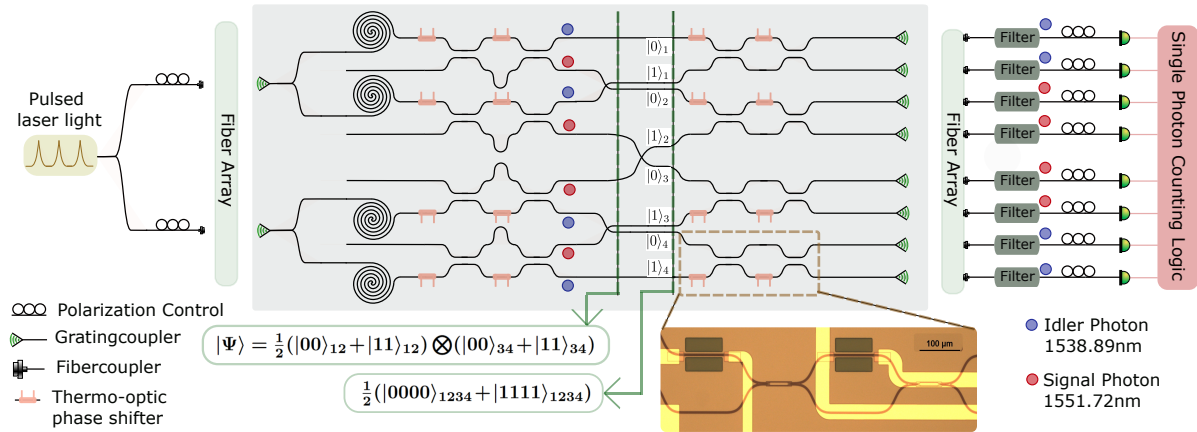


Fig. 2: Experimental setup of the silicon photonic integrated circuit. Photonic chip schematic including quantum state generation, active integrated components, and detection logic. Inset: Image of a Mach-Zehnder Interferometer with two thermo-optic phase shifters and two multimode interferometers used as couplers.

in Figure 1. The task is a binary classification task shown in Fig. 1a (three-dimensional data represented as two-dimensional for ease of illustration) in which training data points, represented as vectors with known labels "+" and "-", are sampled from the distribution presented in [11]. A quantum feature map (Fig. 1b) maps the data into a higher-dimensional feature space, where the kernel entry between these points is the overlap. Here, the quantum circuit (Fig. 1c) is used to sample the kernel entries [12]. The computational resource is a path-encoded four-qubit star graph state (Fig. 1c). By applying local unitary transformations followed by projective measurements, collective three-qubit states can be implemented, resulting in a CBQC register of three qubits [13]. In the presented implementation, the kernel is computed using a measurement-based protocol on a silicon photonic integrated circuit, see Figures 1d and 2. By performing individual single photon measurements, where the projection onto the $|0000\rangle$ and $|1111\rangle$ states are used for calculation, the quantum kernel matrix is reconstructed element-wise (Fig. 1e). Similarly, the matrix for the test data points, where only the vectors are known, is reconstructed. Finally, the reconstructed training and test matrices are fed into a classical SMV. The SVM identifies the optimal separating hyperplane in the feature space (Fig. 1f), which is then used to classify the labels of test data.

Methods

The experimental setup is shown in Figure 2. The pump laser is a pulse train with a 2.5 GHz repetition rate and is filtered to be centered around 1545.32 nm. A 3 dB splitter provides two beam copies that are polarization-aligned and injected into the photonic integrated circuit (PIC) using a fiber-array illuminating grating couplers. Each of the two arms coherently pumps two spiral waveguides, thereby acting as two single-photon pair sources [14]. Each photon pair from a source is

separated by color using an unbalanced Mach-Zehnder interferometer. Then, crossing waveguides realize a fusion gate, generating a four-qubit GHZ state with $n = 4$ photons [13]. Then the photons enter reprogrammable Mach-Zehnder interferometers (MZIs) consisting of an outer and an inner phase shifter that can apply any local unitary $SU(2)$ transformation on each. Here, thermo-optical phase shifters provide a phase range exceeding 2π for full tunability. Each photon is spatially distributed over $m = 2$ modes, resulting in a $m^n = 16$ -dimensional Hilbert space.

A second fiber array collects light from the 8 modes at the PIC output channels and routes it into individual fan-out fibers. Each output port is filtered to ensure spectral purity and remove any remaining pump light. Finally, the light is detected by eight superconducting nanowire single-photon detectors, which feed into an eight-channel counting logic.

The kernel protocol relies on events in which four-qubit states are generated on the chip, equivalent to two sources emitting a single-photon pair simultaneously. In the following, such an event is referred to as a four-fold coincidence. Per four-fold event, only one mode (either the upper or lower mode, $|0\rangle_i$ or $|1\rangle_i$, where $i \in \{1, 2, 3, 4\}$ number of qubits) per qubit is postselected. The postselected states are shown in Figure 3.

Results

In the presented experiment, a path-encoded four-qubit GHZ state $\frac{1}{2}(|0000\rangle + |1111\rangle)$ is created from two Bell pairs $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and a fusion gate on the PIC that is shown in Figure 2. Subsequently, the input data is encoded by applying local unitary qubit operations realized by MZIs.

The pump power is optimized to achieve a 3% generation rate per source, balancing pair generation and multi-photon noise. Two-qubit Bell states are generated on the lower and upper halves of the circuit (see Figure 2), where the preliminary

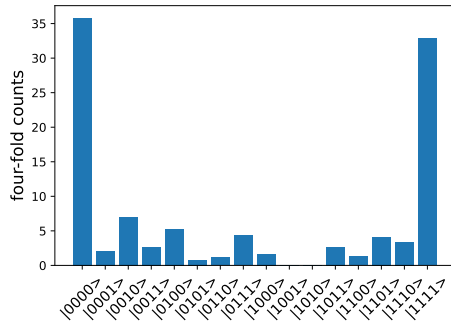


Fig. 3: Measurement of four-qubit Bell state in computational basis, 5 minutes accumulative measurement time, total number of four-fold counts: 105 counts across all 16 channel combinations shown on x-axis

quantum state tomography yields a fidelity of 0.93 for each half.

To obtain the four-qubit state, the PIC is reconfigured into a GHZ state by applying a fusion gate. Each of the four qubits is manipulated through an MZI per qubit that can realize the combined computational gates. The result of setting the four gates in the computational Pauli Z-basis is shown in Figure 3. Here, the single photon count is balanced to $\approx 800 \times 10^3$ counts per second per mode, yielding a total number of 105 four-fold counts across all 16 relevant mode combinations during five minutes of accumulative measurement time. The computational basis measurement has a 97 % fidelity to the target state $\frac{1}{2}(|0000\rangle + |1111\rangle)$. Equally, the GHZ state can be projected in the X-basis $\frac{1}{2}(|+++\rangle + |--\rangle)$. This measurement, which has higher phase sensitivity than the computational basis, yields 71 % accuracy. Here, device performance is limited by the extinction ratio of the MZIs, multi-photon noise, imperfect photon distinguishability, and instabilities in the experimental setup, leading to the accumulation of phase errors as circuit complexity increases and phase shifter crosstalk grows.

To reconstruct the kernel matrices, first a GHZ state is created, and then the kernel mapping is implemented using a combination of local linear optical operations and projective measurements. During the training phase, 30 training data points were sampled from a three-dimensional distribution in which both location and label are known. They are used to generate a symmetric 30×30 matrix, which is shown in Figure 1e. The matrix is reconstructed from 435 individual five-minute measurements, each measurement corresponding to one kernel matrix element on the upper diagonal of the symmetric matrix. Each measurement is performed by reprogramming the PIC with pre-calculated phase settings and accumulating four-fold counts for five minutes. The entry is calculated by analyzing the four-fold output distribution, only using the projected probabilities $p(|0000\rangle) + p(|1111\rangle)$. The element-wise differences between the recon-

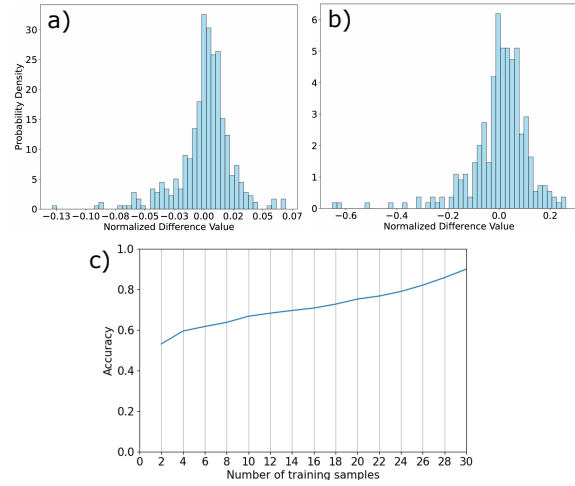


Fig. 4: a) Normalized statistical difference of the training matrix, b) Normalized statistical difference of the testing matrix, c) Classification accuracy with the number of training data points.

structed measurement-based train matrix and the simulated target matrix are shown in Figure 4a. Similarly, the testing matrix, a 30×10 matrix, is constructed from ten data points sampled from the same distribution as the training points, with unknown labels, and reconstructed from 300 measurements. Here, the element-wise differences between the test and target matrix are shown in Figure 4b. Feeding the matrices back into a classical computer to perform the SVM task. Figure 4c shows the training accuracy as a function of the number of training points, indicating that as the number of measured training samples increases, the training efficiency converges towards a training accuracy of 1. Here, training with a larger number of training points can be achieved by increasing the training kernel matrix and the corresponding number of measurements for reconstructing the matrix. Despite hardware-induced inaccuracies in the reconstructed matrices, as seen in Figure 4a and b, the SVM yielded 90 % classification accuracy on the test set, demonstrating the kernel's robustness to instabilities. This makes QKE a feasible approach for measurement-based quantum computing algorithms.

Conclusion

In this work, we proposed and experimentally demonstrated a measurement-based quantum kernel estimation protocol using a reprogrammable silicon PIC using path-encoded four-qubit Bell states as the computational resource. By encoding classical data into local unitary transformations and performing projective measurements, we reconstructed quantum kernel matrices, enabling a classical SVM to achieve 0.9 accuracy on the test dataset. These numbers demonstrate that kernel methods are robust to hardware-induced noise, such as phase drift and multi-photon interference.

Acknowledgments

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